

CS 374A Midterm 1 Review

ACM @ UIUC

September 26, 2026



Disclaimers and Logistics

- **Disclaimer:** We are all current or past CAs, but we have NOT seen the exam. We have no idea what the questions are. However, we've taken the course and reviewed Jeff's previous exams, so we have a good idea of what it'll look like.
- This review session is being recorded. Recordings and slides will be distributed on EdStem after the end.
- **Agenda:** We'll review all topics likely to be covered, then go through a practice exam, then review individual topics by request.
 - Questions are designed to be written in the same style as Jeff's previous exams but to be *slightly* harder, so don't worry if you don't get everything right away!
- Please let us know if we're going too fast/slow, not speaking loud enough/speaking too loud, etc.
- If you have a question anytime during the review session, please ask! Someone else almost surely has a similar question.
- We'll provide a feedback form at the end of the session.

Induction

Template

Let x be an *arbitrary* string/integer/etc.

Inductive Hypothesis: Assume for all k s.t. k is shorter/smaller/etc. than x that $P(k)$ (what we're trying to prove) holds.

Base Case: If $x = 0, \epsilon$, whatever your base case is, then $\dots$, so $P(x)$ holds.

Inductive Step: If $x \neq 0, \epsilon$, whatever your base case is, then $\dots$, so by the inductive hypothesis, $\dots$, so $P(x)$ holds.

Thus, by the principle of induction, $P(x)$ holds.

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Base Case: If $x = 0, \epsilon$, whatever your base case is, then $\dots$, so $P(x)$ holds.

Inductive Step: If $x \neq 0, \epsilon$, whatever your base case is, then $\dots$, so by the inductive hypothesis, $\dots$, so $P(x)$ holds.

Thus, by the principle of induction, $P(x)$ holds.

Some tips:

- Always use strong induction.
- Write out your IH, base case, and inductive step out explicitly.
- Think about what you would like to know about your smaller/shorter numbers/strings.

Regular Languages/Expressions

- Built inductively on 3 operations:

- + is the union operator. $L(r_1 + r_2) = L(r_1) \cup L(r_2)$

- * is the Kleene star. $L(r_1^*) = L(r_1)^*$

- () are used to group expressions

- (implicit) concatenation operator: $L(r_1 r_2) = \{xy : x \in L_1, y \in L_2\}$

~~1~~ * = { ϵ , |, ||, ..., }

0|

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- Closed under Union ($\cup$), intersection ($\cap$), concatenation ($\cdot$), kleene star (*), complement (c), set difference ($\setminus$), and reverse (R)
- ... but only finitely many applications of these operations

$$\{xy^c \mid x, y \in L\}$$

$$L \cdot L^c$$

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 - ... but only finitely many applications of these operations
- If trying to guess whether or not a language is regular, think about memory. DFAs only get finite memory!
 - You don't get to look back indefinitely.
 - If your language requires you to track a number or string indefinitely, it is not regular!

$0^n 1^n$

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 - You don't get to look back indefinitely.
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- **Regex Design Tips:**
 - What strings are in your language? Which ones aren't? Note edge cases (specifically check ϵ).
 - Look for patterns and substrings that you definitely need to include or repeat.

DFAs

- DFA $M = (Q, A, \Sigma, s, \delta)$
 - Q - FINITE set of states
 - $A \subseteq Q$ - accepting states
 - Σ - input alphabet, usually $\{0, 1\}$
 - $s \in Q$ - start state
 - $\delta : Q \times \Sigma \rightarrow Q$ - transition function

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- **Tips for Creating DFAs:**
 - Define your states exactly! What does it mean to be at each state?
 - Based on these definitions, when should you accept? Define A accordingly.
 - What state represents ϵ ? Make that the start state. Make sure that if L accepts ϵ , you accept your start state.
 - How does reading in a 0 or 1 change each state? Define δ accordingly.

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 - How does reading in a 0 or 1 change each state? Define δ accordingly.
- Notes:
 - To go from L to its complement, just switch the accepting and non-accepting states.
 - Every DFA is automatically an NFA.
 - Every regular language can be represented by a DFA. Every DFA represents a regular language.
 - Give states insightful names (e.g. ϵ , 1, 10, 100 not state 1, state 2, ..., state n)

Product Constructions

- Say I can build DFA M_1 keeping track of one property and DFA M_2 keeping track of another property. What if I want a DFA M that keeps track of both properties?
- You can combine the information of both DFAs into one product DFA.
- The accept states in your new DFA define whether I only accept strings with both properties or strings with one or the other or some other logical operation.

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Template

1. Define (drawing/formal) your first DFA $M_1 = (Q_1, A_1, \Sigma, s_1, \delta_1)$.
2. Define (drawing/formal) your second DFA $M_2 = (Q_2, A_2, \Sigma, s_2, \delta_2)$.
3. Define the product DFA $M = (Q, A, \Sigma, s, \delta)$ as follows:
 - ▶ $Q = Q_1 \times Q_2 = \{(q_1, q_2) | q_1 \in Q_1, q_2 \in Q_2\}$ - each state is a tuple
 - ▶ $s = (s_1, s_2)$ - just the tuple containing both start states
 - ▶ $\delta((q_1, q_2), a) = (\delta_1(q_1, a), \delta_2(q_2, a))$ - apply the first transition function to the first state and the second transition function to the second state
 - ▶ $A = \{(q_1, q_2) | q_1 \in A_1 \text{ **and/or/etc.** } q_2 \in A_2\}$ - check if the first state is accepted by the first DFA, check if the second state is accepted by the second DFA, and accept based on the problem statement

NFAs

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- **Tips for Creating NFAs:**
 - Make sure that your transition function consistently leads to a SET of states (whether that set is empty, has 1 state, or multiple states).
 - Use epsilon transitions to jump from one state to another without reading anything (usually when you want to be able to transition from one phase to another).

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- Notes
 - Every NFA can be converted to a DFA through power set construction. The DFA states are the power set of the NFA states.
 - Give states insightful names (e.g. ϵ , 1, 10, 100 not state 1, state 2, ..., state n)

Fooling Sets

- DFAs only care about which state you're in, and not how you got there
 - If two strings result in the same DFA state, any additional suffix added to both will also result in both strings being in the same state.

Regular
 - NFA
 - DFA
 - Regular Eq

Non Regular
 fooling sets
 $F = \{x^i y^j \mid i \geq j\}$
 $x \quad y$

$x \neq y$
 $x \neq y$

Fooling Sets

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 - Thus, if we have x, y , and we know that there exists a **distinguishing suffix** z s.t. $xz \in L, yz \notin L$, then x, y must be in *different* states for *any* DFA that accepts L

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- A **fooling set** is a set of strings where there exists a distinguishing suffix between every pair of strings
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- **Tips and Tricks**
 - If L needs to keep track of a value with no bound, create a fooling set around the part you count up.
 - If you're using strings of the form $1^k, 0^p$, etc. when sampling elements of your fooling set a^i, a^j , you may assume WLOG that $i < j$.

$$F: \{0^n \mid n \geq 1\} \\ 0^i, 0^j, i \neq j, \text{ wlog } i < j$$

Prove $\{0^n 1^n \mid n \geq 0\}$ is irregular.

Template adapted from *Fall 2025 Homework 3 solutions*.

Let $F = \left\{ \boxed{0^n \mid n \geq 0} \right\}$, which is infinite.

Let $x, y \in F$ with $x \neq y$. Thus, $x = \boxed{0^i}$ and $y = \boxed{0^j}$, where $\boxed{\text{wlog } i < j}$.

Let $z = \boxed{1^i}$.

- Then, $xz \in L$ because

$0^i 1^i \rightarrow$ because i is same 0 and 1

- Then, $yz \notin L$ because

$0^j 1^i \rightarrow$ Because $j > i$ $j \neq i$

Thus, z is a distinguishing suffix for x and y , so F is a fooling set for L .

Since F is infinite, L is not a regular language.

Note: We can also pick z so that $xz \notin L$ and $yz \in L$. Just do whichever is easier.

Language Transformations

- We have a language L we know is regular.
- We have a function f from strings to strings: $f(w) = x$.
- We define a transformed language in one of the following formats:
 1. $L' = \{f(w) | w \in L\}$
 2. $L' = \{w | f(w) \in L\}$
- We want to show that L' is regular by making a DFA/NFA that accepts L' .

$$f(w) = w \cdot 0$$

$$x \cdot y \quad f(w) = w^2$$

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Template

- Since L is regular, there is a DFA $M = (Q, A, \Sigma, s, \delta)$ that accepts it.
 - Now, we create a DFA/NFA $M' = (Q', A', \Sigma, s', \delta')$ that accepts L'
- $f(w) = w^R$
 $f(001) = 110$
 $f(011) = 110$
 1011
1. $L' = \{f(w) | w \in L\}$ - M' reads in the results of $f(w) = x$, so we need to undo f to find w from x and then push w through the original DFA M to check acceptance
 2. $L' = \{w | f(w) \in L\}$ - M' reads in strings w , so we need to apply f to w to find $f(w) = x$ and then push x through the original DFA M to check acceptance
- Define the states Q' by thinking about what information you need to interpret the next letter you read - usually $Q \times \{\text{Information needed to convert } x \text{ to } w \text{ or } w \text{ to } x\}$
 - Define the transition function δ so it passes w or x back to the original DFA - potentially using nondeterminism or epsilon transitions

Context-Free Languages/Grammars

- Formally, a context-free grammar is defined by
 - V - nonterminals/variables
 - T - terminals/symbols $\leftarrow \rightsquigarrow$
 - $S \in V$ - start variable
 - P - set of production rules $A \rightarrow \alpha$ with $A \in V$ and $\alpha \in (V \cup T)^*$

$\underline{S} \rightarrow 0S0 \mid 1S0 \mid \dots$
 $\uparrow$
 $\underline{A} \rightarrow$

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- CFLs are only closed under union, Kleene star, and concatenation. CFLs are *not* closed under intersection or complement.

$L_1, L_2 \text{ CFL}$

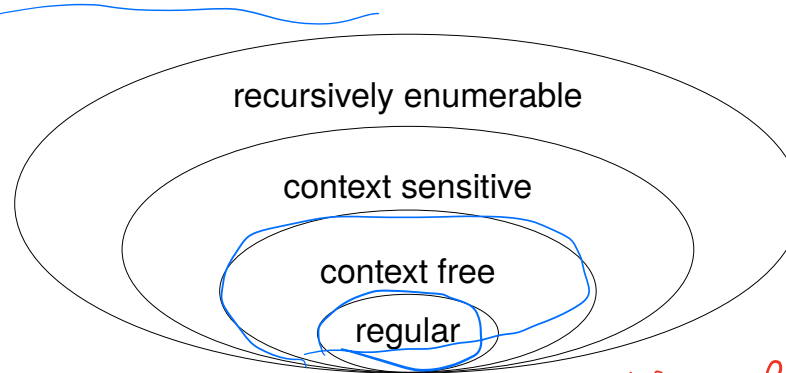
$L_1 \cup L_2 \in \text{CFL}$

$\cap$

$L_1 \cap L_2 \notin \text{CFL}$

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A hand-drawn diagram of a cell membrane. It features a wavy line representing the lipid bilayer, with several small circles inside representing hydrophilic heads and a few vertical lines representing hydrophobic tails. To the right of the bilayer, there are several small circles representing proteins or other molecules. Below the bilayer, there is a single small circle.

if $L_1 \in \mathcal{R}$ and $L_2 \in \mathcal{R}$

L_1 and $L_2 \in \mathcal{R}$
set operation

Short Answer T/F (1)

For each of the following, determine if the statement is **true** or **false**, and give a one-sentence explanation of your answer. (These are intentionally tricky)

- (a) For all languages L , if L is irregular, then L has a finite fooling set.

True $\emptyset$ ∞ ∞

- (b) If M is a minimal DFA that decides a language L , and running M on strings x and y result in states q and q' , respectively, where $q \neq q'$, then there exists a distinguishing suffix between x and y in L .

→ (E) → (fail) $2^{0,1}$ E, fail 0

- (c) The language $L = \{0^i 1^j 0^k : i = j \text{ and } k \equiv i \pmod{374}\}$ is context-free.

$$S \rightarrow s_0 E_0 \mid s_1 E_1 \mid \dots$$

False

Three

- (d) For context-free languages L_1, L_2 , the language $L = (L_1^* L_2) \cup (L_1 L_2^*)$ is context-free.

the

$$L_1^* L_2 \cup \overline{L_1 L_2^*}$$

- (e) (Fall 2024) For all regular languages L_R and context free languages L_C , $L_R \setminus L_C$ is context free.

$$L_2 = \sum^*$$
$$L_R \setminus L_C = L_R \cap L_C$$

False

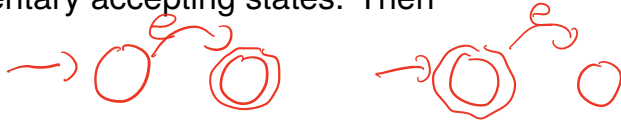
- (f) Suppose $L_1, L_2, \dots$ is an infinite sequence of regular languages, where $L_i \supseteq L_{i+1}$ for all $i \geq 1$. Then, $\bigcup_{i=1}^{\infty} L_i$ is regular.

$$\sum_L^* \rightarrow \sum_{L_2}^0$$

Three Lr

Short Answer T/F (2)

For each of the following, determine if the statement is **true** or **false**, and give a one-sentence explanation of your answer. (These are intentionally tricky)

- (g) The language $\{xx^Ry : x, y \in \{0, 1\}^*\}$ is regular. **True** $X = \epsilon$
 $\{y : y \in \{0, 1\}^*\} = \Sigma^*$
- (h) If L is regular, then $\text{SELFOLD}(L) = \{a_1a_na_2a_{n-1} \cdots a_{\lceil \frac{n}{2} \rceil} : a_1a_2 \cdots a_n \in L\}$ is regular. **True** $(a_1a_2 \cdots a_n a_{n-1} a_{n-2} \cdots a_2 a_1) \text{ odd, even}$
- (i) Consider the language $L = \{1^x2^y3^z : y = x + z\}$. There exists a distinguishing suffix between the strings 1112222223 and 2223. **False** $33 \sim 33$
- (j) Let M_1, M_2 be arbitrary NFAs with identical alphabets, states, starting states, and transition functions, but with complementary accepting states. Then $L(M_1) \cap L(M_2) = \emptyset$. **False** 
- (k) Consider an infinite sequence of regular languages $L_1, L_2, \dots$ s.t. $L_{i-1} \subseteq L_i$. The language $\bigcup_{i=1}^{\infty} L_i$ is context free. **False** $L = \{w_1, w_2, \dots, w_n, \dots\}$
 $L_1 = \{w_1\}, L_2 = \{w_1, w_2\}, \dots$

Regular or Not?

For each of the following languages, either *prove* that the language is regular, or *prove* that it is not regular (**Hint:** exactly two of the four languages are regular). For all questions, $\Sigma = \{0, 1\}$.

- $R \bullet \{1xyx \mid x, y \in \Sigma^*\}$ $x = \epsilon \rightarrow 1y \rightarrow y \in \Sigma^*$
 $NR \bullet \{x1xy \mid x, y \in \Sigma^*\}$
 $R \bullet \{w \in \Sigma^* : |w| \geq 374 \text{ and last 374 characters of } w \text{ have equal number of 0s and 1s}\}$
 $\bullet \text{ (Fall 2021 Conflict) } \{0^p 1^q 0^r \mid r = p + q\}$

$x1xy \rightarrow NR$
 $F = \emptyset$

$0^n 1 \mid n \geq 1$
 $x = 001 \rightarrow 0^i$
 $y = 0001 \rightarrow 0^j$
 $wlog \ i < j$

$z = 0^i$
 $xz \rightarrow \overbrace{001}^x \overbrace{00}^z \rightarrow \in L$
 $yz \rightarrow \overbrace{0001}^y \overbrace{00}^z \rightarrow \notin L$

$\{ \text{ } \}_{374}$

$R = \emptyset$ all strings 374
 and $\#(0 \text{ in } s) = \#(1 \text{ in } s)$

$\hookrightarrow$ Becomes if finite

$0^n 1^n$ non regular $\checkmark$
 $\{0^p 1^q 0^r \mid r = p + q\}$
 $\hookrightarrow (0+1)^* R$

$F = \emptyset$ $0^n \mid n \geq 1$
 $x \rightarrow 0^i$
 $y \rightarrow 0^j$
 $x \neq y$ and $wlog \ i < j$
 $z = 10^{n-1}$

Language Transformations ($\Sigma = \{0, 1\}$)

(Spring 2025) For a language $L \subseteq \Sigma^*$, we define operation DELETEPROPERMID:

$$\text{DELETEPROPERMID}(L) = \{ \overbrace{uw}^{\text{blue}} \mid uvw \in L \text{ and } u, v, w \in \Sigma^*, \underbrace{|v| \geq 2}_{\text{red}} \}$$

Prove that if L is regular, then $\text{DELETEPROPERMID}(L)$ is also regular.

$$v = 101$$

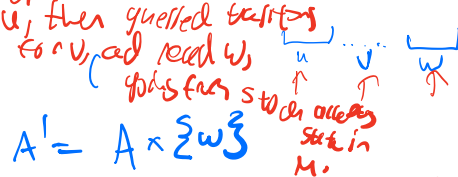
$$w = 11$$

Since L is regular, let $M = (Q, S, \delta, A, \epsilon)$ be a DFA for L .

Given: $0011 \in$

Accept: 001011

if M accepts a string then it would have read u , then guessed v and read w .



We will define an NFA with

$$\epsilon\text{-transition) } M' = (Q', S', \delta', A')$$

$$A' = A \times \{w\}$$

$$Q' = Q \times \{u, 1, 2t, w\} \leftarrow (q,$$

$$S' = (s, u)$$

Assume all unspecified transitions to $\emptyset$

$$S': \delta'((q, u), a) = \{(\delta(q, a), u)\}$$

$$\delta'((q, w), a) = \{(\delta(q, a), w)\}$$

$$\delta'((q, u), \epsilon) = \{(\delta(q, 0), 1), (\delta(q, 1), 1)\}$$

$$\delta'((q, 1), \epsilon) = \{(\delta(q, 0), 2t), (\delta(q, 1), 2t)\}$$

Language Transformations ($\Sigma = \{0, 1\}$)

(Fall 2024) For a string w , let $\text{swap}(w)$ denote the set of all strings formed by selecting any number of disjoint pairs of consecutive bits in w , and then swapping the bits within each pair. For example, $\text{swap}(01) = \{01, 10\}$ and

$$\text{swap}(0101) = \{0101, \underline{10}01, 00\underline{11}, 01\underline{10}, \underline{1010}\}$$

For a language L , let $\text{SWAP}(L) = \bigcup_{w \in L} \text{swap}(w)$. Prove that if L is regular, so is $\text{SWAP}(L)$.

Language Transformations ($\Sigma = \{0, 1\}$)

(Spring 2025 Homework) We define an operation DELETEONES that removes all 1s in a string. For example,

$$\text{DELETEONES}(101010010001) = 0000000$$

Prove that for a regular language L ,

$$\text{DELETEONES}(L) = \{\text{DELETEONES}(w) \mid w \in L\}$$

is also regular.

DFAs/NFAs/Regexes

With $\Sigma = \{0, 1\}$,

(a) Write a regex for strings with no even-length runs.

DFAs/NFAs/Regexes

With $\Sigma = \{0, 1\}$,

- (a) Write a regex for strings with no even-length runs.
- (b) Write a DFA *and* regex for $\{w \in \Sigma^* \mid \#_0(w) \geq 2 \text{ or } \#_1(w) \geq 2\}$.

DFAs/NFAs/Regexes

With $\Sigma = \{0, 1\}$,

- (a) Write a regex for strings with no even-length runs.
- (b) Write a DFA *and* regex for $\{w \in \Sigma^* \mid \#_0(w) \geq 2 \text{ or } \#_1(w) \geq 2\}$.
- (c) All strings that do not contain 010 as a substring.

CFGs

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(a) $\{ww^R : w \in \{0, 1\}^* \text{ and } |ww^R| \equiv 1 \pmod{3}\}$

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- (c) $\{0^x 1^y 2^z : x - y = z\}$

Feedback



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